MUHAMMAD UMAIR HAROON

EPLAN Electric P8 Engineer · LV Switchgear & Control Panel Design

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Karachi, Pakistan · UTC+5 · Remote

EXPERIENCE	4+ years
CORE TOOL	EPLAN Electric P8
LARGEST SET	90 sheets
UP TO	5550 A · 100 kA
ENGAGEMENT	Freelance / remote

A1 PROFILE

Electrical design engineer with 4+ years producing complete EPLAN Electric P8 documentation for low-voltage switchgear and control panels. My packages run from plant characteristics and general arrangements through power and control schematics, PLC/controller I/O, terminal diagrams and manufacturer-coded bills of material. I have delivered synchronizing, ATS, MCC/EMCC and distribution panel sets of up to 90 sheets, rated up to 5550 A / 100 kA, for airport, hospitality and industrial projects in Saudi Arabia, and I currently work remotely for a UK engineering firm on protection and control schematics. I work comfortably to a client's own title blocks and standards, turn revisions around quickly, and can carry a panel from first issue to as-built on my own.

A2 CORE SKILLS FOR THIS ROLE

EPLAN Electric P8: multi-page schematics, device tagging and cross-referencing, wire numbering, parts management, automatic reports (ToC, terminal diagrams, BOM), revision control

PLC and I/O: digital/analog I/O wiring for DEIF AGC 150 controllers and ALC-4 load-shed PLCs, HMI, Modbus RTU (RS-485) and Ethernet networks

Terminal and wiring documentation: terminal line-up diagrams, external/internal connections, ferrule schedules

Control panel design: power and control circuits, interlocking, ACB/MCCB control (motor operator, closing coil, shunt trip, UVR)

Relay and MCC drawings: interface relays, DOL, star-delta, VFD and dual-speed starters, earth-leakage relays, BMS signals

Panel layout and BOM: front, component and busbar layouts (EPLAN and AutoCAD); BOMs with ABB, Eaton, Phoenix Contact, DEIF and Lovato order codes

B1 EXPERIENCE

Electrical Design Engineer, Protection & Control (Remote)

Sep 2024 – Present

UK-based engineering firm · working remotely from Karachi

- Prepare AC and DC secondary schematics for protection and control panels covering busbar, differential, restricted earth fault and distance protection schemes.
- Develop trip, close and interlocking circuits and relay wiring for 33/11 kV and 400/33 kV substation projects, coordinating with the client's secondary design team.
- Configure and parameterise protection relays in Siemens DIGSI, Siemens Reyrolle Reydisp and ABB PCM600.
- Projects include the Eggborough 400/33 kV BESS and Rowden 33/11 kV primary substation.

Electrical Design Engineer, LV Switchgear & Control Panels

May 2022 – Sep 2024

ACDC Electrical Services, Karachi · technical department, panel projects for the Middle East

- Produced complete EPLAN Electric P8 drawing packages for MDBs, SMDBs, DBs, MCC/EMCC, ATS and LV synchronizing panels, from requirement analysis through to as-built issue.
- Each package covered plant characteristics, GA and component layouts, power and control schematics, controller I/O, terminal line-up diagrams and BOM. The largest set ran to 90 sheets.
- Designed generator synchronizing and load-sharing schemes around DEIF AGC 150 controllers, with ALC-4 load-shed PLCs, a 15" AGI HMI, Modbus RTU (RS-485) and an industrial Ethernet network.
- Developed HVAC motor control circuits with DOL, star-delta, VFD and dual-speed starters, earth-leakage protection and BMS status and command signals.
- Detailed breaker control for Eaton and ABB ACBs/MCCBs: motor operators, closing coils, shunt trips, undervoltage releases and mechanical/electrical interlocks.
- Built BOMs with manufacturer order codes and managed revisions through for-approval, IFC and as-built issues, folding in consultant and client comments (11 revisions on one panel).

- Drew general arrangements and busbar layouts in EPLAN and AutoCAD for line-ups up to 4.2 m, working to IEC 61439 forms of separation, IEC 60364 and NEK 400.

C1 SELECTED EPLAN PROJECTS

PROJECT	SCOPE	SET
Jizan Airport DMCP, KSA *Synchronization control system*	Seven generators and a bus tie on Bus A/B, 8 × DEIF AGC 150, 2 × ALC-4 load-shed PLC, AGI 415 HMI, 16-port Ethernet switch, Modbus RTU, signal interface to 13.8 kV MV switchgear, 66 interface relays	90 sheets Rev 05
Addoha Poultry, Middle East *LV synchronizing panel*	5550 A, 400 V 60 Hz, 100 kA (1 s) busbar, Form 2B, 4.1 m five-section line-up, Eaton IZMX40 2500 A motorised ACBs, 4 × DEIF AGC 150	50 sheets As-built R11
Shura Island HC-02 / HC-03 Hotels, KSA *HVAC MCC / EMCC*	DOL, star-delta, VFD and dual-speed starters, earth-leakage relays, BMS interface, Form 3B Type 2, IP54; plus MDB, SMDB, DB, ATS and synch panels	Multiple panels
ATS panel 1000 A *Automatic transfer switch*	480/277 V 60 Hz, IP65, ABB XT7 1000 A motorised MCCBs with Ekip trip units, Lovato ATL601 controller, mechanical interlock	29 sheets

D1 SOFTWARE

EPLAN Electric P8 (daily tool) · AutoCAD · Siemens DIGSI · Reydisp · ABB PCM600 · Microsoft Office

D2 STANDARDS

IEC 61439 (LV switchgear assemblies, forms of separation) · IEC 60364 · NEK 400

E1 EDUCATION

B.E. Electrical Engineering
NED University of Engineering & Technology, Karachi
Oct 2018 – Oct 2022

E2 LANGUAGES

English (professional)

Drawing samples from these projects are available in my online portfolio (link in my cover email) or on request.